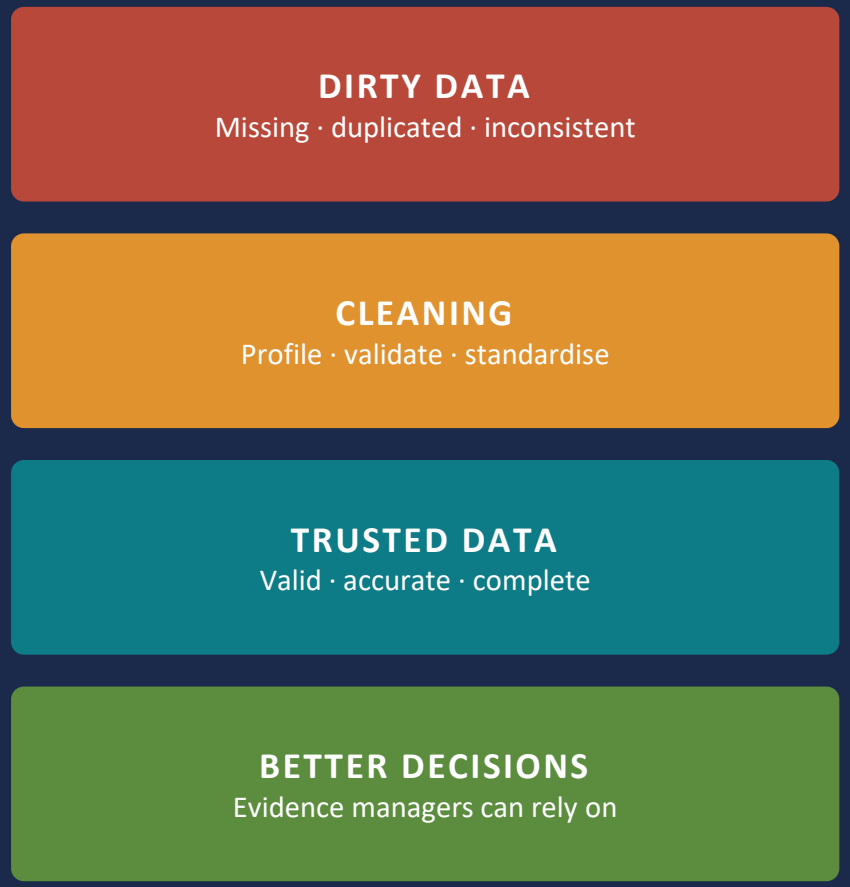
DATA SCIENCE FOR BUSINESS · CHAPTER 3

Data Quality and Data Preparation

Missing data · Duplicates · Inconsistent formats · Outliers · Quality dimensions · Cleaning concepts · Data lineage

Why poor-quality data produces poor business decisions

Data Science Fundamentals for Business School Students



Where Chapter 3 Fits

Chapter 2 recap

We learned to turn a vague problem into a **measurable analytical question**.

We chose the **unit of analysis, target variable and data sources**.

We compared internal/external and primary/secondary data.

Chapter 3 focus

Before analysis can begin, we must ask:

- Is this data **fit for purpose**?
- What is **wrong** with it?
- How do we **prepare** it responsibly?
- Can we **trace** where it came from?

Why it matters

A perfect question and a sophisticated model cannot rescue bad data.

Analysts typically spend **60–80% of project time** finding and fixing data problems.

Big message Data quality is a **business risk**, not a technical detail. A decision made on duplicated customers, stale prices or mis-typed dates can cost more than having no analysis at all — because people **trust** a number that looks official.

BY THE END OF THIS CHAPTER YOU WILL BE ABLE TO

Learning Objectives

1. **Explain** why poor-quality data produces poor business decisions.
2. **Identify** missing data, duplicates, inconsistent formats and outliers in a dataset.
3. **Apply** the six data-quality dimensions: validity, accuracy, completeness, consistency, timeliness, uniqueness.
4. **Evaluate** options for handling missing values, duplicates and outliers, and justify your choice.
5. **Design** a simple, documented data-cleaning plan for a business dataset.
6. **Describe** data lineage and why traceability matters for trust, audit and compliance.

Assessment alignment You will be assessed on **spotting problems, choosing a treatment and justifying it** — the justification carries as many marks as the fix.

LECTURE STRUCTURE

Today's Roadmap



PART 1

Why quality matters

Garbage in, garbage out — and what it costs.



PART 2

Quality dimensions

Validity, accuracy, completeness, consistency, timeliness, uniqueness.



PART 3

Common problems

Missing data, duplicates, inconsistent formats, outliers.



PART 4

Cleaning concepts

Profiling, validation, standardisation, imputation, documentation.



PART 5

Data lineage

Where data came from and what happened to it.



PART 6

Practice & assessment

Scenarios, challenges, knowledge checks and a test.

01

PART 1

Why Data Quality Matters

Garbage in, garbage out — and what it costs the business



IN THIS PART

- Warm-up: spot the problems
- Garbage in, garbage out
- Business consequences
- The 1–10–100 rule
- Where problems come from

Warm-up: What Is Wrong With This Table?

In pairs, list every data-quality problem you can see. How many can you find in two minutes?

Customer ID	Name	Join date	Age	Region	Spend	Email
C001	Sara Ahmed	12/03/2024	31	North	£240.00	sara@mail.com
C002	J. Smith	2024-04-02	45	South	£180.50	jsmith@mail.com
C003	Lola Diaz	05/05/2024	220	north	£310.00	lola@mail.com
C004	Omar Khan	19/05/2024	N/A	East	95 EUR	(blank)
C005	John Smith	May 19 2024	45	South	£180.50	jsmith@mail.com
C006	Mia Rossi	21/05/2024	28	West	unknown	mia@mail.com



Discuss Which of these problems would a manager notice in a report — and which would stay invisible but still change the answer?

Garbage In, Garbage Out



Why errors grow: Aggregation

One wrong row changes a total, an average and a growth rate.

Why errors grow: Automation

An error copied into a daily report repeats every day until someone checks.

Why errors grow: False confidence

A clean-looking chart hides the mess behind it — nobody questions it.

Principle No model is better than the data it was trained on.

Business Consequences of Poor Data

Wrong decisions

A region looks unprofitable only because its sales were recorded in the wrong currency.

Lost revenue

Marketing mails the same customer three times and misses another entirely.

Compliance and legal risk

Inaccurate customer records breach data-protection duties and regulatory reporting.

Wasted analyst time

Teams rebuild the same fixes every month instead of doing analysis.

Operational failures

Wrong addresses, failed deliveries, stock-outs and angry customers.

Lost trust in data

Once a report is wrong twice, managers stop using it and decide on instinct.

Note Poor data rarely announces itself. The analysis still runs, the chart still looks professional — only the **conclusion** is wrong.

The Cost of Poor-Quality Data

£1

Prevention

Validate at entry: dropdowns, required fields, format checks.

£10

Correction

Find and fix it later: profiling, cleaning, re-running reports.

£100

Failure

Live with it: wrong decisions, lost customers, regulatory penalties.



The 1–10–100 rule The later a data error is caught, the more it costs. **Prevention is always cheaper than correction, and correction is always cheaper than failure.**



Discuss Where in your organisation (or part-time job) is data still typed in by hand with no validation?

PART 1 · BUSINESS SCENARIO · 8 MINUTES

Scenario 1 — The Duplicate Customer Problem

Situation A retailer reports **480,000 customers**. An audit finds that **60,000 records are duplicates** created when customers checked out as guests with slightly different name spellings. Marketing has just approved a retention campaign costing **£0.40 per customer contacted**.

Which metrics are wrong?

Think about customer count, revenue per customer, churn rate and campaign ROI.

What does it cost?

Estimate the wasted campaign spend and the cost of annoying customers with repeat mail.

How would you fix it?

What matching rule would you use, and what would you change at the point of entry?

Deliverable Three bullet points: the metric most distorted, your estimate of the cost, and one prevention control.

Where Do Data Quality Problems Come From?

Manual entry

Typos, free-text fields, rushed staff, customers entering their own details.

System integration

Two systems merged with different keys, field names and formats.

Migration

Data moved to a new system; fields truncated, mapped wrongly or lost.

Decay over time

People move, change jobs, change names; prices and products change.

Definition mismatch

“Active customer” means something different to sales, finance and marketing.

Process gaps

No validation rules, no owner, no routine checks on the data.

Insight Most data problems are **organisational, not technical** — so the fix usually involves process and ownership, not only code.

Knowledge Check 1 — Symptom and Cause

Symptom seen in the report	Likely cause	Business impact
Customer count rose 14% overnight with no campaign		
Average order value fell sharply in one region only		
A fifth of loyalty members have no age recorded		
Last month’s sales still missing from the dashboard		



Pair discussion For one row, name the **control** that would have prevented the problem at source.

Knowledge Check 1 — Symptom and Cause

Symptom seen in the report	Likely cause	Business impact
Customer count rose 14% overnight with no campaign	**Duplicate records after a system merge**	Per-customer metrics understated
Average order value fell sharply in one region only		
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A fifth of loyalty members have no age recorded	**Optional free-text field at sign-up**	Segmentation excludes those customers
Last month’s sales still missing from the dashboard	**Late data feed — a timeliness failure**	Decisions made on stale information



Pair discussion For one row, name the **control** that would have prevented the problem at source.

02

PART 2

Dimensions of Data Quality

A shared vocabulary for describing what is wrong

IN THIS PART

- Validity and accuracy
- Completeness and consistency
- Timeliness and uniqueness
- Measuring quality with KPIs
- Audit activity and knowledge check

The Six Dimensions of Data Quality

Validity

Does the value follow the rules and allowed format?

Accuracy

Does the value match reality?

Completeness

Is anything missing — fields, rows or periods?

Consistency

Does it agree across systems and over time?

Timeliness

Is it recent enough for this decision?

Uniqueness

Is each real-world thing recorded only once?

Why a framework helps "The data is bad" tells nobody what to do. **"Completeness of the age field is 68%"** tells the business exactly what to fix, and who owns it.

Dimension 1 — Validity

Validity means

The value **follows the defined rules**: correct data type, format, range and allowed list of values.

Validity asks "**could** this value exist?", not "is it true?"

Examples of failure

- Date of birth: 31/02/2024
- Age: 220 or −5
- Email with no "@"
- Region: "Nrth" (not on the approved list)
- Postcode in a phone-number field

How to check it

Apply validation rules: type checks, range checks, format patterns and lookup lists.

Business impact

Invalid values break calculations, crash imports and force analysts to exclude rows.

Dimension 2 — Accuracy

Accuracy means

The value **matches reality** — the real person, product, transaction or event it describes.

A value can be perfectly valid and still be wrong.

Examples of failure

- A real but outdated address
- Price recorded as £19.99 when it was £1.99
- Wrong customer linked to an order
- Meter reading typed one digit out

How to check it

Verify against an independent source, sample-audit records, or confirm with the customer.

Business impact

Inaccurate data sends deliveries to the wrong place and prices products wrongly.

Dimension 3 — Completeness

Completeness means

Nothing required is missing — at value level, row level and period level.

Ask: missing **fields**, missing **records**, or missing **time periods**?

Examples of failure

- 32% of customers have no age recorded
- One store did not upload last week's sales
- A product category exists in the catalogue but never in the sales table

How to check it

Measure the percentage of non-missing values per field, and count rows per period.

Business impact

Gaps bias averages, shrink samples and can make a bad month look like a good one.

Dimension 4 — Consistency

Consistency means

The data **agrees with itself** across systems, across fields and over time.

Includes definition consistency: the same term must mean the same thing everywhere.

Examples of failure

- CRM says 12,400 customers; finance says 11,980
- Order date later than delivery date
- “Active customer” defined differently by sales and finance
- Units in kg in one feed and lb in another

How to check it

Reconcile totals between systems and run cross-field rules (e.g., delivery date ≥ order date).

Business impact

Inconsistency starts arguments in meetings and destroys confidence in every number.

Dimension 5 — Timeliness

Timeliness means

The data is **available and up to date enough for the decision** being made.

Two useful terms: **latency** (delay between event and availability) and **freshness** (age of the newest record).

Examples of failure

- Stock levels updated once a night for a same-day delivery promise
- Prices refreshed weekly in a market that moves hourly
- A dashboard still showing last quarter

How to check it

Compare the age of the newest record with the decision cycle it supports.

Business impact

Stale data causes over-selling, missed price moves and decisions about a world that has changed.

Dimension 6 — Uniqueness

Uniqueness means

Each real-world entity is recorded **once and only once**.

Closely linked to **integrity**: keys and relationships between tables hold.

Examples of failure

- The same customer stored as “J. Smith” and “John Smith”
- An order imported twice after a failed upload
- A product with two different codes

How to check it

Count distinct keys versus total rows, and run fuzzy matching on names, emails and addresses.

Business impact

Duplicates inflate customer counts, deflate per-customer metrics and waste campaign budget.

Comparison Table — The Six Dimensions

Dimension	Question it answers	Example failure	Typical check
Validity	Could this value exist?	Date of birth 31/02/2024	Type, range, format, allowed list
Accuracy	Is it true?	Price recorded as £19.99 instead of £1.99	Verify against an independent source
Completeness	Is anything missing?	32% of customers have no age	% non-missing per field; rows per period
Consistency	Does it agree everywhere?	CRM and finance report different totals	Reconciliation and cross-field rules
Timeliness	Is it recent enough?	Stock refreshed nightly for same-day orders	Age of newest record vs decision cycle
Uniqueness	Is it recorded only once?	“J. Smith” and “John Smith” are one person	Distinct keys vs row count; fuzzy matching



Exam tip **Validity = follows the rules. Accuracy = matches reality.** A value can be valid and still completely wrong.

Measuring Data Quality: Simple KPIs

KPI	How it is calculated	Business example
Completeness rate	Non-missing values ÷ total rows	% of customers with a valid email
Error / invalid rate	Failed validation rules ÷ total rows	% of rows failing the date-format rule
Duplicate rate	(Rows − distinct keys) ÷ rows	% of customer records that are duplicates
Freshness lag	Today − date of newest record	Hours since the last sales upload
Reconciliation gap	|System A total − System B total|	Difference between CRM and finance counts
Accuracy sample rate	Correct records in an audited sample	% correct in a 100-record manual check



Set thresholds Agree a **target and an alert level** per field (e.g., email completeness ≥ 98%, freshness lag ≤ 24 hours) and assign an **owner** to each.

PART 2 · GROUP ACTIVITY · 10 MINUTES

Activity 1 — Audit a Dataset

Use the warm-up table, or a dataset you have found. Score each dimension and give evidence.

Dimension	Score (good / fair / poor)	Evidence from the data	Business risk if ignored
Validity			
Accuracy			
Completeness			
Consistency			
Timeliness			
Uniqueness			



Report back Which **single dimension** would you fix first, and what evidence justifies putting it ahead of the others?

Knowledge Check 2 — Name the Dimension

Case	Which dimension failed?	Answer
A delivery date is recorded as 31 November		**Validity**
The same supplier appears under two different IDs		**Uniqueness**
Today's dashboard still shows last month's sales		**Timeliness**
Marketing counts 9,800 active users; finance counts 8,400		**Consistency**
A quarter of orders have no delivery postcode		**Completeness**
A customer's recorded address is a real address — but not theirs		**Accuracy**



Discuss case 6 Why can this error **not** be detected inside the dataset itself? What external check would reveal it?

03

PART 3

Common Data Problems

Missing data, duplicates, inconsistent formats and outliers

IN THIS PART

- Missing data and why it happens
- Options for handling gaps
- Duplicates and deduplication
- Inconsistent formats
- Outliers: detect and treat

Missing Data: How It Hides

Truly blank

An empty cell or NULL in the database.

Risk: Easy to count — most tools flag it automatically.

Placeholder text

“N/A”, “unknown”, “none”, “-”, “TBC”.

Risk: Looks like data; breaks numeric columns.

Sentinel numbers

999, −1, 0 used to mean “not recorded”.

Risk: Silently included in averages and totals.

Default dates

01/01/1900 or today’s date auto-filled.

Risk: Distorts tenure, age and duration calculations.

First step Before analysing any column, list its **distinct values and extremes**. Hidden placeholders are usually obvious the moment you look at the minimum, maximum and most frequent values.

Why Data Goes Missing — and Why It Matters

Missing completely at random

Example: A scanner failed for an hour; the gap has nothing to do with the customer.

Impact: Least harmful — mostly a loss of sample size.

Missing for a related reason

Example: Younger customers skip the income field more often than older ones.

Impact: Analysis becomes biased towards the groups who do answer.

Missing because of the value itself

Example: The highest earners are the least likely to disclose their income.

Impact: Most dangerous — the gap hides exactly the values you need.

Key idea The **pattern** of missing data matters more than the amount. Ask: *are the records with gaps different from the ones without?* If yes, deleting them introduces bias into every result.

Options for Handling Missing Data

Option	What you do	When it is reasonable	Risk
Investigate first	Find out why the values are missing	Always — before any other option	Takes time, but prevents bad choices
Leave as missing	Keep the gap and report it	When the tool handles gaps honestly	Different charts use different row counts
Drop the rows	Remove incomplete records	Few rows affected and missing at random	Bias and a smaller sample if not random
Drop the field	Remove a mostly empty column	Field is >60–70% empty and not essential	You lose a variable that may matter
Impute a value	Fill with mean, median, mode or a model	Gaps are moderate and patterns are stable	Invents data; shrinks real variation
Flag and keep	Add a “value was missing” indicator	Missingness itself may be informative	Adds a column; needs explaining
Collect it again	Go back to the source	High-value decisions; small samples	Slowest and most expensive option



Golden rule Whatever you choose, **document it and report how many values were affected.** A silent fix is a hidden assumption.

PART 3 · BUSINESS SCENARIO · 8 MINUTES

Scenario 2 — The Missing Income Field

Situation A bank builds a loan-default model on 40,000 applications. **Income is missing for 22%** of applicants. An analyst deletes those rows to "keep the data clean". Later it emerges that **self-employed applicants** were far more likely to leave income blank — and they default at a different rate.

What bias was introduced?

Who disappeared from the training data, and how does that change the model?

What were the alternatives?

Consider flagging, imputation, a separate model, or collecting the field again.

What is the fairness risk?

What happens to self-employed applicants when this model is used to decide?

Deliverable A three-sentence recommendation to the risk committee: what you would do instead, and what you would disclose.

Duplicates: Exact and Fuzzy

All four rows below are the same customer. Only the first pair is an exact duplicate.

ID	Name	Email	Phone	City	Type
C100	John Smith	j.smith@mail.com	07700 900111	Leeds	Original
C101	John Smith	j.smith@mail.com	07700 900111	Leeds	**Exact duplicate**
C214	J. Smith	j.smith@mail.com	07700900111	Leeds	**Fuzzy duplicate**
C488	Jon Smyth	jsmith@mail.com	+44 7700 900111	Leeds	**Fuzzy duplicate**

Why duplicates are dangerous

They inflate counts, deflate per-customer averages, double-count revenue and waste contact budget — all without producing any obvious error.

Why they appear

Guest checkouts, re-registrations, system merges, failed uploads re-run, and free-text name fields with no unique key.

Prevention A **verified unique identifier** at sign-up (email or phone) prevents most duplicates before they exist.

Deduplication Concepts

Matching key

The field (or combination) used to decide whether two records are the same thing: email, phone, or name + postcode + date of birth.

Matching rule

Exact match, or fuzzy match with a similarity threshold that tolerates typos and spelling variants.

Survivorship rule

Which value wins when records disagree: the most recent, the most complete, or the most trusted source.

Golden record

The single merged record that the business agrees to treat as the truth for that customer or product.

Caution Over-matching merges **two different people** — a worse error than leaving a duplicate. Review borderline matches manually and **never delete the original records**.

Inconsistent Formats

Field	What you find in the raw data	Standardised form
Dates	12/03/2024 · 2024-03-12 · 12 Mar 24 · 03/12/2024	ISO format: 2024-03-12
Currency	£240 · 240 GBP · 240.00 · “two hundred forty”	Numeric 240.00 with a separate currency field
Text case / spacing	“north” · “North” · “ NORTH ”	Trimmed, consistent case: “North”
Units	kg and lb, minutes and hours, mixed in one column	One unit per column, stated in the data dictionary
Phone numbers	07700900111 · +44 7700 900111 · (0)7700-900111	One canonical pattern, e.g. +447700900111
Categories	“M”, “Male”, “male”, “1”	A fixed, documented list of allowed values



The date trap **03/04/2024** is 3 April in the UK and 4 March in the US. Without knowing the source system, no tool can tell — which is why ISO dates (YYYY-MM-DD) are the professional standard.

Standardisation Concepts

Canonical format

Agree one target format per field (ISO dates, GBP to two decimals, lower-case emails) and convert everything to it.

Lookup / reference table

Map every observed variant to the approved value: “north”, “Nrth”, “N” all map to “North”.

Validation at entry

Dropdowns, required fields, input masks and format checks stop bad values being created at all.

Data dictionary

Document each field: meaning, type, allowed values, units and owner — so future analysts do not re-invent the rules.

Remember Cleaning fixes **today’s** dataset; validation at entry and a shared data dictionary fix **every future dataset**.

Outliers: What They Are

Definition

An outlier is a value that lies **far away from the rest of the data** for that variable — unusually high, unusually low, or an impossible combination of values.

Why they matter

A single extreme value can move a mean, a trend line and a forecast — while the median barely changes.

Data-entry error
£10,000 typed instead of £100.00

Measurement or system fault
A sensor logs 0 while offline

Rare but genuine event
A corporate order 50× the usual size

A different population
Wholesale buyers inside a retail dataset

Key idea An outlier is a **question, not automatically an error**. Investigate before you touch it.

Detecting Outliers

Range and business rules

Check the minimum and maximum against what is possible: age 0–120, delivery time > 0, discount ≤ 100%.

Visual inspection

Histograms, box plots and scatter plots make extreme values obvious in seconds.

Interquartile range (IQR)

Flag values below Q1 − 1.5×IQR or above Q3 + 1.5×IQR — the standard box-plot rule.

Standard deviations (z-score)

Flag values more than about 3 standard deviations from the mean (assumes a roughly symmetric distribution).

Context matters An order of £8,000 is an outlier among consumers and completely normal among business customers. **Always check within the relevant segment.**

Treating Outliers: A Decision Flow

1. Detect
Flag the value with rules, charts or statistics.

2. Investigate
Check the source record. Is it possible? Is it plausible?

3. Decide
Correct · keep · cap · exclude — with a reason.

4. Document
Record what you did and how many rows were affected.

Correct it

The true value is known or recoverable from the source.

Keep it

It is genuine — report it and consider the median alongside the mean.

Cap it (winsorise)

Limit extreme values to a stated percentile — and disclose that you did.

Exclude it

Only when it is clearly an error, and only with documentation.

Never Never delete values simply because they **spoil the result you wanted**. That is manipulation, not cleaning.

Other Problems Worth Knowing

Wrong data type

Numbers stored as text, so they sort as 1, 10, 2 and cannot be summed.

Truncation

Long names or codes cut off during export, creating false duplicates.

Encoding problems

Accented characters turn into symbols after a bad file transfer.

Structural problems

Merged cells, subtotal rows inside the data, headers repeated mid-file.

Mixed granularity

Daily rows and monthly summary rows stored in the same table.

Broken relationships

Orders referring to a customer ID that no longer exists.

Practical habit Open every new file and check: row count, column types, first and last rows, and the minimum and maximum of each numeric field.

Knowledge Check 3 — Problem and Treatment

What you observe	Problem type	Defensible treatment
18% of rows have no postcode	**Completeness / missing data**	Investigate the cause, flag and keep; drop only if random
“J. Smith” and “John Smith”, same email	**Uniqueness / fuzzy duplicate**	Match on email, merge with a survivorship rule, keep originals
Dates appear in three formats	**Validity / inconsistent format**	Convert all to ISO, then validate against the source system
One order of £82,000 from a business buyer	**Genuine outlier**	Keep it, report it separately, use the median alongside the mean
Age recorded as 999 for 400 customers	**Sentinel value hiding a gap**	Convert 999 to missing, then choose a missing-data strategy



Stretch For each row, name **one prevention control** that would stop the problem being created in the first place.

PART 3 · GROUP ACTIVITY · 12 MINUTES

Activity 2 — Clean This Table

Order ID	Customer	Order date	Age	Region	Amount	Email
ORD-101	Ana Perez	04/02/2024	34	North	£120.00	ana@mail.com
ORD-102	Ken Ito	2024-02-05	0	north	£89.99	ken@mail.com
ORD-103	Ana Perez	07/02/2024	34	North	150 USD	ana@mail.com
ORD-104	Leo Moro	09/02/2024	41	South	£64.00	(blank)
ORD-104	Leo Moro	09/02/2024	41	South	£64.00	(blank)
ORD-105	Zoe Li	Feb 11 2024	181	West	£7,400.00	zoe@mail.com

Your task — produce a cleaning plan, not just a tidy table:

1. List issues
2. Name dimension
3. State the fix
4. State assumption
5. Prevention control

Watch out Is the £7,400 order an error or a genuine high-value customer? How would you find out?

04

PART 4

Data Cleaning and Preparation Concepts

A repeatable, documented process — not ad-hoc edits



IN THIS PART

- The preparation workflow
- Profiling and validation rules
- Cleaning operations
- Golden rules and reproducibility
- The ethical line

The Data Preparation Workflow

1

Profile

Understand what you actually have: counts, types, ranges, gaps.

2

Validate

Test the data against rules and flag every failure.

3

Clean

Standardise, deduplicate, correct, handle gaps and outliers.

4

Transform

Derive variables, aggregate, join and reshape for analysis.

5

Document

Record every decision so the work can be repeated and audited.

It is a loop, not a line

Cleaning reveals new problems, so you profile again. Two or three passes are normal.

Documentation is continuous

Record decisions as you make them — not at the end, when you have forgotten why.

Professional standard The workflow should be a **script that can be re-run**, not a set of manual edits nobody can reproduce.

Data Profiling: What to Check First

Size and shape

How many rows and columns? Does the row count match what you expected?

Types per column

Is each column the type it should be — number, date, text, category?

Missing values

What percentage of each field is missing, and is it concentrated anywhere?

Ranges and extremes

Minimum, maximum, mean and median for every numeric field.

Distinct values

How many categories exist, and are there obvious variants of the same one?

Keys and duplicates

Is the ID unique? How many rows share the same key?

Mindset Profiling is **diagnosis before treatment**. Never clean a column before you understand what is in it and why.

Validation Rules

Rule type	What it checks	Business example
Type rule	The value is the right kind of data	Order amount must be numeric, not text
Range rule	The value sits within sensible limits	Age between 16 and 110; discount 0–100%
Format rule	The value follows a required pattern	Email contains “@”; postcode matches the national pattern
Allowed-list rule	The value is one of the approved options	Region is one of: North, South, East, West
Cross-field rule	Two fields agree with each other	Delivery date is on or after the order date
Referential rule	A key exists in the related table	Every order’s customer ID exists in the customer table

Good practice Write validation rules **once**, run them on every refresh, and report the **number of failures per rule** — that is a data-quality dashboard.

Core Cleaning Operations

Standardise

Convert formats, units, cases and category labels to one agreed form.

Deduplicate

Identify and merge records that describe the same real-world entity.

Handle missing values

Leave, flag, impute, drop or re-collect — with the reason recorded.

Correct errors

Fix values that can be verified against a reliable source.

Filter

Remove rows that are out of scope — test records, cancelled orders, other periods.

Derive and enrich

Create new variables: age from date of birth, tenure, totals, segments.

Remember Every operation on this slide **changes the data**. Each one needs a reason, a record, and a count of the rows affected.

Cleaning vs Transformation

Cleaning — making data correct

- Fixing invalid and impossible values
- Standardising formats and categories
- Removing duplicates
- Handling missing values

Goal: the data faithfully represents reality.

Transformation — making data usable

- Deriving variables (age, tenure, margin)
- Aggregating to the right unit of analysis
- Joining tables together
- Binning values into bands or segments

Goal: the data fits the question being asked.

Why the distinction matters Cleaning is **the same for everyone** who uses the dataset; transformation is **specific to one analysis**. Keep them in separate, documented steps so others can reuse your clean data with a different question.

Golden Rules of Data Cleaning

1. Never overwrite the raw data — always work on a copy.
2. Keep a cleaning log: what you changed, why, and how many rows were affected.
3. Make it reproducible: a script or recorded steps, not manual edits.
4. Investigate before you fix — understand the cause, not just the symptom.
5. State your assumptions in the final report, not only in your notes.
6. Re-profile after cleaning to confirm you fixed it and broke nothing else.



The test Could another analyst take your raw file and your documentation, repeat every step, and get **exactly the same numbers**? If not, the work is not finished.

The Ethical Line: Cleaning vs Manipulating

Legitimate cleaning

- Removing a value that is **impossible** (age 220)
- Correcting a value against a **verified source**
- Excluding **test or cancelled** records, and saying so
- Capping extremes with a **stated, disclosed** rule

Manipulation

- Deleting genuine low scores to lift an average
- Dropping the worst-performing region “because it distorts the trend”
- Trying several cleaning rules and keeping the one with the best result
- Not reporting how many rows were removed

The test Would you be comfortable if the **full cleaning log were published** alongside your conclusion? Legitimate decisions are made on evidence and **disclosed**; manipulation is chosen for its effect and **hidden**.

PART 4 · BUSINESS SCENARIO · 8 MINUTES

Scenario 3 — The Convenient Cleaning Rule

Situation A pilot store refit is being evaluated. The first analysis shows **no sales uplift**. An analyst notices that two refitted stores had unusually low weekend sales, calls them "data errors caused by till problems", excludes them — and the uplift becomes **+4.2%**. The board paper is due tomorrow.

Is the exclusion defensible?

What evidence would make it legitimate — and what would make it manipulation?

What must be disclosed?

What should appear in the board paper alongside the +4.2% figure?

What would you do?

You are the junior analyst. What do you say, and to whom?

Principle to apply Decide the cleaning rule **before** you see its effect on the answer — and apply it to every store, not only the inconvenient ones.

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PART 5

Data Lineage and Governance

Where did this number come from — and who is responsible for it?



IN THIS PART

- What data lineage is
- Why traceability matters
- Metadata and data dictionaries
- Roles and ownership
- Reproducibility

What Is Data Lineage?

Definition

Data lineage is the **documented journey of data**: where it originated, every transformation applied to it, and every report or model it feeds.

It answers: **“Where did this number come from, and what happened to it on the way?”**

Source
POS system

Extract
Nightly export

Clean
Dedupe, standardise

Transform
Join, aggregate

Report
Board dashboard

The weak link The step most often undocumented is the **analyst’s own spreadsheet**: manual edits, filters and pasted values that nobody can reconstruct six months later.

Why Lineage Matters to the Business

Audit and compliance

Regulators ask you to prove where a reported figure came from.

Trust

People believe a number more when its journey can be explained.

Debugging

When a figure looks wrong, lineage shows where to look first.

Impact analysis

If a source field changes, you know instantly which reports break.

Handover

New analysts can take over without reverse-engineering old files.

Reproducibility

The same inputs and steps produce the same numbers again.

The board question "Where did this number come from?" If nobody can answer within a few minutes, the organisation has a **lineage problem**, not a reporting problem.

Metadata and the Data Dictionary

A data dictionary records what each field means — and who is responsible for it.

Field	Meaning	Type / allowed values	Source	Owner
customer_id	Unique customer identifier	Text, pattern C###	CRM	CRM team
join_date	Date account was created	Date, ISO format	CRM	CRM team
region	Sales region of the billing address	North / South / East / West	Billing system	Finance
active_flag	Purchased in the last 90 days	Yes / No, refreshed nightly	Derived	Analytics



The argument it prevents Sales says 9,800 active customers; finance says 8,400. Both are right — they used **different definitions of "active"**. A dictionary makes the definition explicit and owned.

Who Is Responsible? Data Governance Roles

Data owner

A senior business person accountable for a data domain (customer, product, finance): sets the definitions and the quality targets.

Data steward

Day-to-day custodian of quality: monitors the KPIs, investigates failures and maintains the dictionary.

Data custodian / IT

Runs the systems that store and move the data: access, security, backups, pipelines.

Data consumer

Analysts and managers who use the data: responsible for reporting problems rather than quietly working around them.

Governance in one line Every important field needs a **named owner**. Unowned data stays dirty.

Reproducibility: Script, Don’t Click

Manual cleaning

- Edits typed directly into the file
- Nobody can see what changed
- Must be redone by hand every month
- Impossible to audit or hand over

Scripted cleaning

- Steps written once and re-run on demand
- Every change is visible and reviewable
- Same inputs always give the same outputs
- Easy to audit, share and improve

A minimum cleaning log — usable even without code

Step	Field	Action taken	Rows affected	Reason / assumption
1	order_date	Converted three formats to ISO	1,240	Source system confirmed day-first format
2	customer_id	Merged fuzzy duplicates on verified email	318	Email is a verified unique identifier
3	age	Converted sentinel value 999 to missing	402	999 used by the old system to mean “not given”

Knowledge Check 4 — Lineage and Governance

Question	Your answer	Model answer
What does data lineage record?		**Origin, every transformation, and every report the data feeds**
Who is accountable for the definition of “active customer”?		**The data owner for the customer domain**
What artefact records field meanings and allowed values?		**The data dictionary (metadata)**
An analyst spots a broken field. What should they do?		**Report it to the steward — not quietly patch their own copy**
Why prefer a script to manual edits?		**Reproducible, auditable, reviewable and re-runnable**

Discuss Who should pay for fixing a data-quality problem: the team that **created** it, or the team that **discovered** it?

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PART 6

Practical, Challenges and Assessment

Applying data quality thinking to realistic business situations

IN THIS PART

- Data-quality canvas
- Cleaning-plan activity
- Three graded challenges
- Scenario pack and test
- Key takeaways

Practical Canvas — Data Quality Assessment

Canvas item	Your answer
1. Dataset and its source system	
2. Decision this data will support	
3. Fields that must be correct for that decision	
4. Quality issues found (by dimension)	
5. Treatment chosen for each issue	
6. Assumptions introduced by those treatments	
7. Prevention control for next time	
8. Owner of the data going forward	
9. What you will disclose in the final report	



How to use it

Complete the canvas **before** you analyse anything.

Item 3 keeps the work proportionate: not every field needs to be perfect — only the ones the decision depends on.

Most often forgotten: items 6, 7 and 8.

PART 6 · GROUP ACTIVITY · 15 MINUTES

Practical Activity — Build a Cleaning Plan

1

Profile

Row counts, types, missing percentages, ranges, distinct values, duplicate keys.

2

List the issues

For each one, name the dimension it breaks and the rows affected.

3

Choose treatments

State the fix, the assumption it makes and the risk it carries.

4

Document and prevent

Write the cleaning log and name one control to stop it recurring.

Deliverable A **one-page cleaning plan**: issues found, dimension affected, treatment chosen, assumption made, rows affected, and one prevention control.

Challenge 1 — Name the Problem

EASY

For each observation, name the problem, the dimension it breaks, and one way to fix it.

Observation in the data	Problem	Dimension	Fix
Dates stored as 12/03/2024 and 2024-03-12 in one column			
The same supplier appears twice with different IDs			
Age is recorded as 999 for 400 customers			
One transaction is 50 times larger than any other			
Delivery date is earlier than the order date			
The sales table has no rows for the last nine days			



Stretch Rank the six issues by **business risk** for a retailer planning next month's stock, and justify your top two.

Challenge 2 — Choose a Missing-Data Strategy

MEDIUM

Case	Your strategy	Justification	Assumption you are making
A. 4% of delivery times missing, spread evenly across all stores			
B. 30% of income values missing, mostly from self-employed applicants			
C. 85% of the "preferred contact time" field is empty			
D. 12 of 60 interviewed customers skipped the satisfaction question			

Stretch For case **B**, explain who is harmed if the rows are simply deleted — and what you would tell the risk committee about the model's limitations.

Challenge 3 — Full Data Quality Audit

HARD

Situation A university wants to predict which first-year students need academic support. Data comes from **four systems**: student records, the attendance app, the virtual learning environment and the library. Each uses a different student identifier, two are updated weekly and one in real time, and attendance is missing for seminars taught off-campus.

Required output	Your answer
Three quality risks you expect, by dimension	
How you would join four systems with different identifiers	
Your missing-data strategy for off-campus attendance	
Two validation rules you would implement	
Your lineage and documentation plan	
One ethical risk and how you would manage it	

Scenario Pack — Four Mini Cases

Logistics

A courier reports 98% on-time delivery, but customers complain constantly. Delivery timestamps are written by drivers at the end of their shift.

Healthcare

A clinic's waiting-time report excludes patients with no recorded arrival time — mostly those who arrived during the busiest hours.

Finance

Two systems report different revenue for the same month. One counts the order date, the other the payment date.

E-commerce

Average session value tripled overnight. A tracking update started recording each page view as a separate session.

For each scenario Name the **dimension at risk**, the **business consequence**, your **treatment**, and one **prevention control**.

ASSESSMENT · MULTIPLE CHOICE

End-of-Lecture Test — Part A

Question	A	B	C	D
1. A date of 31/02/2024 fails which dimension?	Validity	Accuracy	Timeliness	Uniqueness
2. The same customer stored twice fails:	Completeness	Uniqueness	Validity	Timeliness
3. Age recorded as 999 is best described as:	A genuine outlier	A sentinel value hiding a gap	A duplicate	A format error
4. The safest first step with an outlier is to:	Delete it	Cap it	Investigate it	Ignore it
5. Data lineage mainly records:	Staff access rights	Data’s origin and transformations	Server costs	Chart formatting
6. Cheapest point to fix a data error is:	At entry (prevention)	During cleaning	In the report	After the decision

Instructions Work individually for six minutes, then compare with a partner before the answers are revealed.

ASSESSMENT · SHORT ANSWER

End-of-Lecture Test — Part B

1. Explain the difference between **validity** and **accuracy**, with one business example of each.
2. Give two reasons why deleting all rows with missing values can be dangerous.
3. Describe two ways to detect duplicate customer records, and one risk of over-matching.
4. What is data lineage, and why would an auditor ask for it?
5. A colleague removes the two worst-performing stores because they “distort the trend”. What would you say, and what should be disclosed?

Submission format 3–5 bullet points per answer. Marks are for **justification and disclosure**, not for length.

ASSESSMENT

Answer Key and Marking Rubric

Part A answers

1. A — Validity

2. B — Uniqueness

3. B — Sentinel value hiding a gap

4. C — Investigate it

5. B — Origin and transformations

6. A — At entry (prevention)

Rubric for Part B — full credit requires

- **Q1:** validity = follows the rules; accuracy = matches reality; one example of each
- **Q2:** loss of sample size **and** bias if the gaps are not random
- **Q3:** two methods (exact key, fuzzy matching) plus the risk of merging two real people
- **Q4:** lineage defined as origin + transformations + destinations; audit needs traceability
- **Q5:** challenge the decision, require evidence and consistency, decide the rule before seeing the effect, and disclose the exclusion and its impact on the result

SUMMARY

Key Takeaways

1. Poor data quality produces **confidently wrong decisions** — the analysis still runs.
2. Six dimensions give a shared vocabulary: **validity, accuracy, completeness, consistency, timeliness, uniqueness**.
3. The four classic problems are **missing data, duplicates, inconsistent formats and outliers**.
4. Always **investigate before you fix** — and never delete a value just because it is inconvenient.
5. Cleaning must be **documented and reproducible**; raw data is never overwritten.
6. **Lineage and ownership** make numbers trustworthy; prevention at entry is the cheapest control.

Rule to remember **No analysis is better than the data behind it — and no data is trustworthy unless you can say where it came from.**

NEXT STEPS

Preparation for Chapter 4

Before the next lecture

- Profile one real dataset: rows, types, missing %, duplicates.
- Write a short cleaning log for three issues you find.
- Bring one example of a data problem you have met at work or university.
- Confirm your Python / Jupyter access for the lab.

Next chapter: descriptive statistics

- Mean, median and mode — and when each misleads
- Spread, variation and distributions
- Outliers revisited, with statistics
- Communicating summaries clearly to managers

Closing prompt Think of a number your organisation reports every month. **Could you trace it back to its source?**